MEDIUM

ADVANCED MATH

Advanced Math

09

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

Which expression is equivalent to $\sqrt[7]{x^9y^9}$, where x and y are positive?

A. $xy^{\frac{7}{9}}$

B. $xy^{\frac{9}{7}}$

C. xy^{16}

D. xy^{63}

x	hx
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x, since Hana started training in pole vault, and the estimated height hx, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

A. $hx = 1.120.23^x$

B. $hx = 1.121.23^x$

C. $hx = 1.230.12^x$

D. $hx = 1.231.12^x$

Q3

ADVANCED MATH

$$w^2 + 12w - 40 = 0$$

Which of the following is a solution to the given equation?

A. $6 - 2\sqrt{19}$

B. $2\sqrt{19}$

C. $\sqrt{19}$

D. $-6 + 2\sqrt{19}$

Q4 ADVANCED MATH

If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

A. $3x^2 + 12x + 7$

B. $3x^2 + 14x + 17$

C. $3x^2 + 19x + 20$

D. $3x^2 + 26x + 33$

$$h(t) = -16t^2 + 110t + 72$$

The function above models the height h, in feet, of an object above ground t seconds after being launched straight up in the air. What does the number 72 represent in the function?

A. The initial height, in feet, of the object

B. The maximum height, in feet, of the object

C. The initial speed, in feet per second, of the object

D. The maximum speed, in feet per second, of the object

Q6 ADVANCED MATH

$$6x^2 + 5x - 7 = 0$$

What are the solutions to the given equation?

A. $\dfrac{-5 \pm \sqrt{25 + 168}}{12}$

B. $\dfrac{-6 \pm \sqrt{25 + 168}}{12}$

C. $\dfrac{-5 \pm \sqrt{36 - 168}}{12}$

D. $\dfrac{-6 \pm \sqrt{36 - 168}}{12}$

Q7 ADVANCED MATH

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

A. $x7x^{10} - 7x^9 + 87x$

B. $x8^{10} - 8^9 + 88$

C. $8xx^{10} - x^9 + 11x$

D. $8xx^9 - x^8 + 11$

Q8

ADVANCED MATH

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x?

A. 5

B. 10

C. 18

D. 36

Q9 ADVANCED MATH

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b, x, and y. Which equation correctly expresses x in terms of b and y?

A. $x = \frac{7by}{11}$

B. $x = y - 77b$

C. $x = \frac{y}{77b}$

D. $x = 77by$

Q10

GRID-IN ADVANCED MATH

$$\left(\frac{1}{2}x+\frac{3}{2}\right)\left(\frac{3}{2}x+\frac{1}{2}\right)$$

The expression above is equivalent to ax^2+bx+c, where a, b, and c are constants. What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.